

Technical Portfolio Documentation

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Professional Profile & Skills

Highly motivated technical professional transitioning into **Cybersecurity and Software Engineering**. Proven ability to build complex monitoring pipelines, automate data workflows, and implement defensive security measures.

Core Certifications

- **Cybersecurity Certificate:** Foundational and practical knowledge in network defense and threat mitigation.
- **SQL Certificate:** Advanced data querying, database management, and relational mapping.

Technical Toolkit

Kali Purple

Parrot OS

Elasticsearch

Grafana

Snort 3

Filebeat

Python

Network Monitoring

Featured Project: Real-Time IDS Flow Visualizer

Overview

Developed a live Intrusion Detection System (IDS) dashboard to transform raw Snort 3 logs into actionable security intelligence. This project addresses the challenge of "log fatigue" by providing visual context to network traffic.

The Architecture

- **Ingestion:** Configured Snort 3 to generate alerts in an isolated environment.
- **Pipeline:** Used Filebeat to ship logs to an Elasticsearch backend.
- **Visualization:** Built a custom Grafana dashboard featuring a **Sankey Flow Panel**.

Technical Challenges Overcome

- **Data Merging:** Resolved "Not Found" errors by auditing Elasticsearch index mappings to identify correct IP field paths.
- **Advanced Logic:** Implemented **Binary Operation Transformations** to merge source and destination event counts into a single "MainStat" for accurate volume visualization.
- **Threat Intelligence:** Integrated **VirusTotal Data Links**, allowing analysts to perform instant IP reputation checks directly from the visualization.

The Outcome

A dynamic, high-visibility dashboard that categorizes traffic by **Origin / Attacker IP** and **Target / Asset IP**, utilizing color thresholds to highlight high-volume threats instantly.

Defensive Lab Operations

Utilizing **Kali Purple** for defensive security operations, including network auditing, vulnerability assessment, and laboratory-based penetration testing. Experience includes managing security containers (Docker) and configuring Parrot OS for secure development environments.

Collaboration & AI Integration

This project workflow was optimized using **Google Gemini** as a technical assistant. Gemini assisted in debugging complex Grafana transformations, auditing JSON structures, and architecting the data pipeline logic, demonstrating an ability to leverage cutting-edge AI tools to accelerate project delivery and solve complex engineering roadblocks.
